

CHIAYI AP, Taiwan

WMO: 467460

Lat: 23.462N

Lon: 120.393E

Elev: 26

StdP: 101.01

Time Zone: 8.00 (E08)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.6	10.0	5.2	5.5	10.1	7.1	6.3	11.3	9.8	17.4	8.6	17.3	2.5	10	0.343

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	34.0	27.8	33.2	27.5	32.9	27.4	28.8	32.6	28.2	32.1	27.9	31.7	4.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	24.0	31.5	27.2	23.0	30.9	26.9	22.6	30.6	93.8	32.6	91.4	32.3	89.5	31.8	31.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.4	6.5		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
				2.1	30.2	1.9	0.8	0.8	30.8	-0.3	31.3	-1.4	31.7	-2.7	32.3
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.7	17.4	17.8	20.0	23.7	26.4	28.2	29.0	28.7	27.6	25.2	22.0	18.5	
	DBStd	4.85	2.66	3.07	3.50	3.14	1.98	1.58	1.11	1.18	1.55	1.89	2.58	2.67	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	156	50	43	24	4	0	0	0	0	0	0	4	30	
	CDD10.0	5014	228	218	311	411	508	547	588	580	527	471	361	265	
	CDD18.3	2128	19	27	77	166	250	297	329	322	277	212	115	36	
	CDH23.3	19682	86	129	417	1169	2222	3289	3867	3624	2717	1453	557	150	
	CDH26.7	7261	9	20	102	368	787	1332	1627	1440	982	464	116	14	
Wind	WSAvg	2.7	3.2	3.1	3.0	2.6	2.3	2.8	2.6	2.6	2.3	2.4	2.6	2.7	
Precipitation	PrecAvg	1967	27	52	55	110	172	369	420	438	223	49	27	24	
	PrecMax	3230	90	328	159	457	406	916	960	1168	1026	327	211	95	
	PrecMin	1028	0	0	4	0	9	23	61	98	41	0	0	0	
	PrecStd	565	22	72	48	105	98	225	240	304	231	74	45	25	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.0	29.0	31.1	33.1	33.8	34.9	34.4	34.2	33.6	32.2	30.4	28.2	
		MCWB	22.1	22.9	24.6	26.4	27.4	28.0	27.8	28.1	27.6	26.5	24.9	23.0	
	2%	DB	25.8	26.9	29.2	31.5	32.2	34.0	33.9	33.4	32.9	31.1	29.2	26.5	
		MCWB	20.1	21.5	23.7	25.6	26.7	27.8	27.8	27.9	27.5	26.1	24.2	21.4	
	5%	DB	23.9	24.9	27.8	30.2	31.8	33.1	33.2	33.0	32.0	30.2	28.0	25.0	
		MCWB	19.0	20.5	23.0	25.0	26.5	27.4	27.5	27.7	27.1	25.5	23.5	20.4	
	10%	DB	22.1	22.9	25.9	28.9	30.8	32.1	32.8	32.1	31.2	29.3	26.7	23.8	
		MCWB	18.1	19.3	21.9	24.4	26.1	27.0	27.4	27.4	26.8	24.9	22.6	19.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.5	23.6	25.4	26.9	28.2	29.2	29.3	29.6	28.8	27.8	26.0	23.7
			MCDB	26.9	28.1	29.9	31.6	32.0	33.3	32.9	32.7	32.2	31.2	29.4	27.6
2%		WB	20.8	22.1	24.1	26.0	27.5	28.3	28.6	28.7	28.1	26.6	24.9	22.1	
		MCDB	24.7	25.9	28.5	30.6	31.3	32.5	32.5	32.4	31.7	30.1	28.6	25.5	
5%		WB	19.5	20.9	23.1	25.4	26.7	27.8	28.0	28.2	27.5	26.0	24.0	20.9	
		MCDB	22.9	23.9	26.9	29.6	30.6	31.9	32.1	31.8	31.1	29.6	27.1	24.1	
10%		WB	18.5	19.6	22.2	24.6	26.2	27.2	27.6	27.7	27.0	25.3	23.0	19.8	
		MCDB	21.5	22.6	25.3	28.3	29.7	31.0	31.4	31.2	30.4	28.7	25.7	22.8	
Mean Daily Temperature Range	5% DB	MDBR	9.9	9.3	9.2	8.7	7.9	7.5	7.4	7.4	7.7	9.0	9.5	10.6	
		MCDBR	11.8	11.4	11.7	10.3	8.9	8.7	8.4	8.2	8.3	9.3	10.1	11.7	
		MCWBR	7.2	6.5	7.0	5.6	4.8	4.3	4.1	4.3	4.3	5.1	6.1	7.2	
	5% WB	MCDBR	10.7	10.7	10.8	9.8	8.3	8.1	7.9	7.7	7.9	9.0	9.6	10.7	
		MCWBR	6.8	6.6	6.7	5.7	4.7	4.4	4.2	4.5	4.4	5.3	6.1	7.0	
Clear-Sky Solar Irradiance	taub	0.509	0.563	0.691	0.693	0.586	0.529	0.540	0.597	0.628	0.610	0.532	0.522		
	taud	1.969	1.866	1.620	1.665	1.938	2.109	2.072	1.900	1.803	1.796	1.973	1.955		
	Ebn at Noon	739	728	658	669	743	779	771	729	695	679	712	707		
	Edh at Noon	165	195	261	254	192	161	167	198	214	206	164	163		
All-Sky Solar Radiation	RadAvg	3.38	3.80	4.12	4.42	4.81	5.07	5.53	4.98	4.85	4.51	3.58	3.29		
	RadStd	0.37	0.47	0.43	0.55	0.59	0.53	0.40	0.52	0.39	0.28	0.30	0.33		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)		+0.28	N/A	N/A	N/A	+0.18	+0.25	N/A	N/A	+121	+111

Nomenclature: See separate page